



ANGELES UNIVERSITY FOUNDATION

Angeles City

College of Computer Studies

Center of Excellence in IT Education

SAM10 Final Project

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Course: Information Technology

Final Project SAM10

Book Tracker

The screenshot shows a web browser with multiple tabs. The active tab is titled "Book Tracker" and shows a URL starting with "natkimbooktracker-dnargdh4anghd9ee.southeastasia-01.azurewebsites.net". The browser's address bar and tabs are visible at the top. The main content area of the web application has a blue header with the text "My Book Tracker". Below the header, there is a form on the left side titled "Add New Book". This form contains three input fields: "Title" with the text "The Book of SAM10", "Author" with the text "Kim Sangil", and "Status" with a dropdown menu showing "Reading". Below these fields is a blue button labeled "Add Book". To the right of the form is a table with three columns: "TITLE", "AUTHOR", and "STATUS". The table is currently empty, displaying the text "No books yet." in the center.

Link:

<https://natkimbooktracker-dnargdh4anghd9ee.southeastasia-01.azurewebsites.net/>

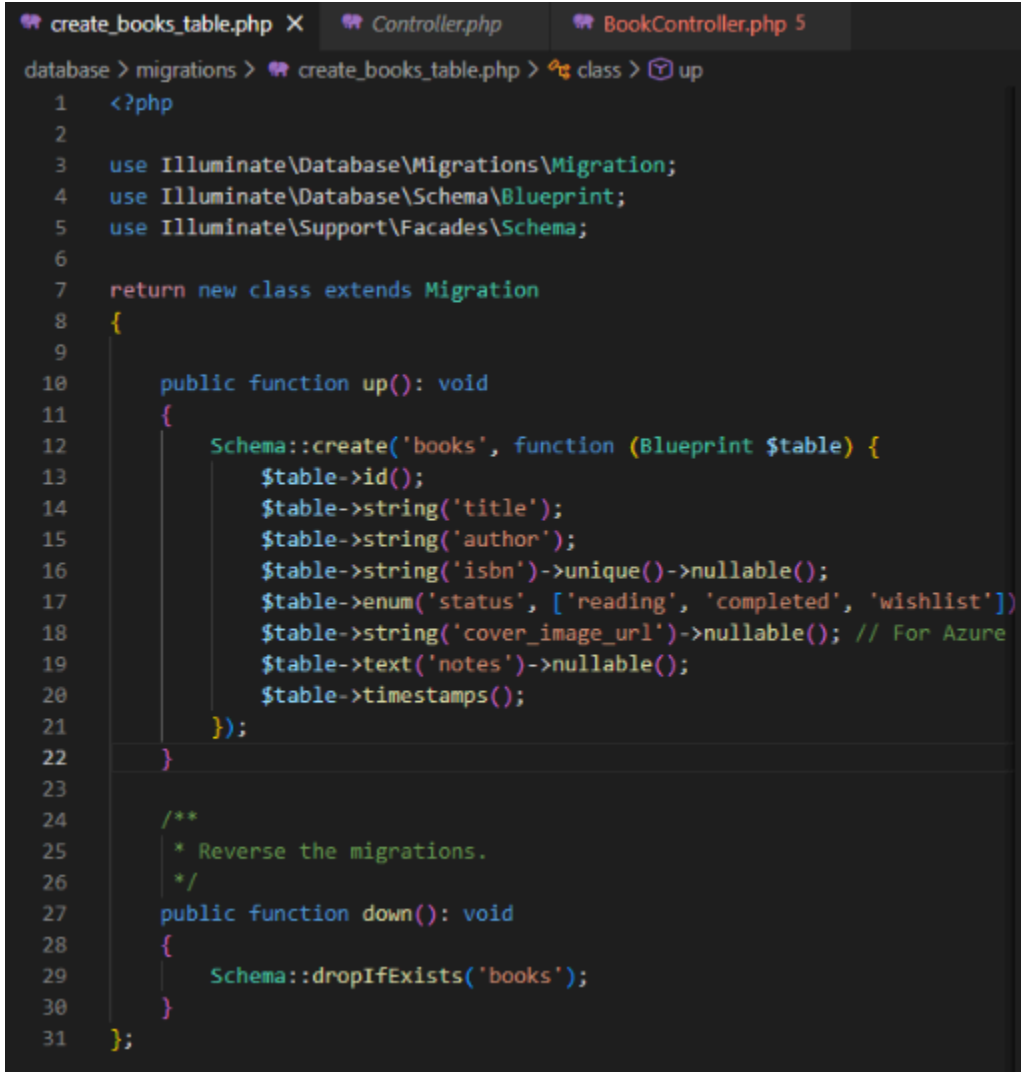
1: Create Laravel Project

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> composer create-project laravel/laravel NatKim_BookTracker
Creating a "laravel/laravel" project at "./NatKim_BookTracker"
Installing laravel/laravel (v12.10.1)
- Installing laravel/laravel (v12.10.1): Extracting archive
Created project in C:\Windows\System32\NatKim_BookTracker
> @php -r "file_exists('.env') || copy('.env.example', '.env');"
Loading composer repositories with package information
Updating dependencies
Lock file operations: 111 installs, 0 updates, 0 removals
- Locking brick/math (0.14.0)
```

2: Create migrations file

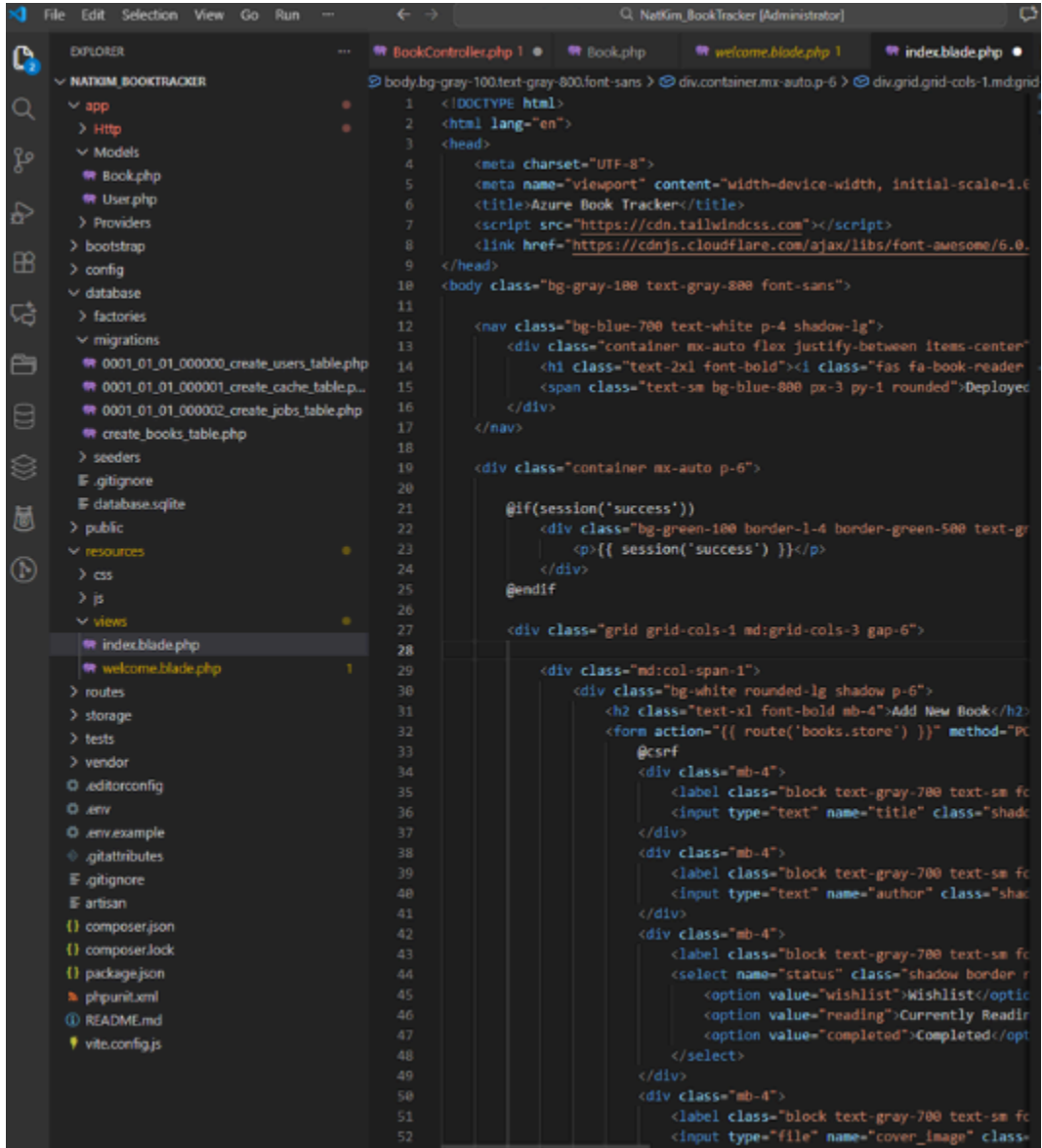


```
database > migrations > create_books_table.php > class > up
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration
8  {
9
10     public function up(): void
11     {
12         Schema::create('books', function (Blueprint $table) {
13             $table->id();
14             $table->string('title');
15             $table->string('author');
16             $table->string('isbn')->unique()->nullable();
17             $table->enum('status', ['reading', 'completed', 'wishlist']);
18             $table->string('cover_image_url')->nullable(); // For Azure
19             $table->text('notes')->nullable();
20             $table->timestamps();
21         });
22     }
23
24     /**
25      * Reverse the migrations.
26      */
27     public function down(): void
28     {
29         Schema::dropIfExists('books');
30     }
31 };
```

3: Create controller file

```
create_books_table.php  Controller.php  BookController.php 5
app > Http > Controllers > BookController.php > BookController
1  <?php
2
3  namespace App\Http\Controllers;
4
5  use App\Models\Book;
6  use Illuminate\Http\Request;
7  use Illuminate\Support\Facades\Storage;
8
9
10
11 class BookController extends Controller
12 {
13     public function index()
14     {
15
16         $books = Book::orderBy('updated_at', 'desc')->get();
17         return view('books.index', compact('books'));
18     }
19
20     public function store(Request $request)
21     {
22         $validated = $request->validate([
23             'title' => 'required|max:255',
24             'author' => 'required|max:255',
25             'status' => 'required|in:reading,completed,wishlist',
26             'cover_image' => 'nullable|image|max:2048'
27         ]);
28
29         $book = new Book($validated);
30
31
32         if ($request->hasFile('cover_image')) {
33             $path = $request->file('cover_image')->store('covers', 'azure');
34             $book->cover_image_url = Storage::disk('azure')->url($path);
35         }
36
37         $book->save();
38
39         return redirect()->route('books.index')->with('success', 'Book s
40     }
41
42     public function update(Request $request, Book $book)
43     {
44         $validated = $request->validate([
45             'status' => 'required|in:reading,completed,wishlist',
46             'notes' => 'nullable|string'
47         ]);
48
49         $book->update($validated);
50
51         return redirect()->route('books.index')->with('success', 'Book s
52     }
53
54     public function destroy(Book $book)
55     {
56         $book->delete();
57         return redirect()->route('books.index')->with('success', 'Book r
58     }
```

4: Create index.blade.php file



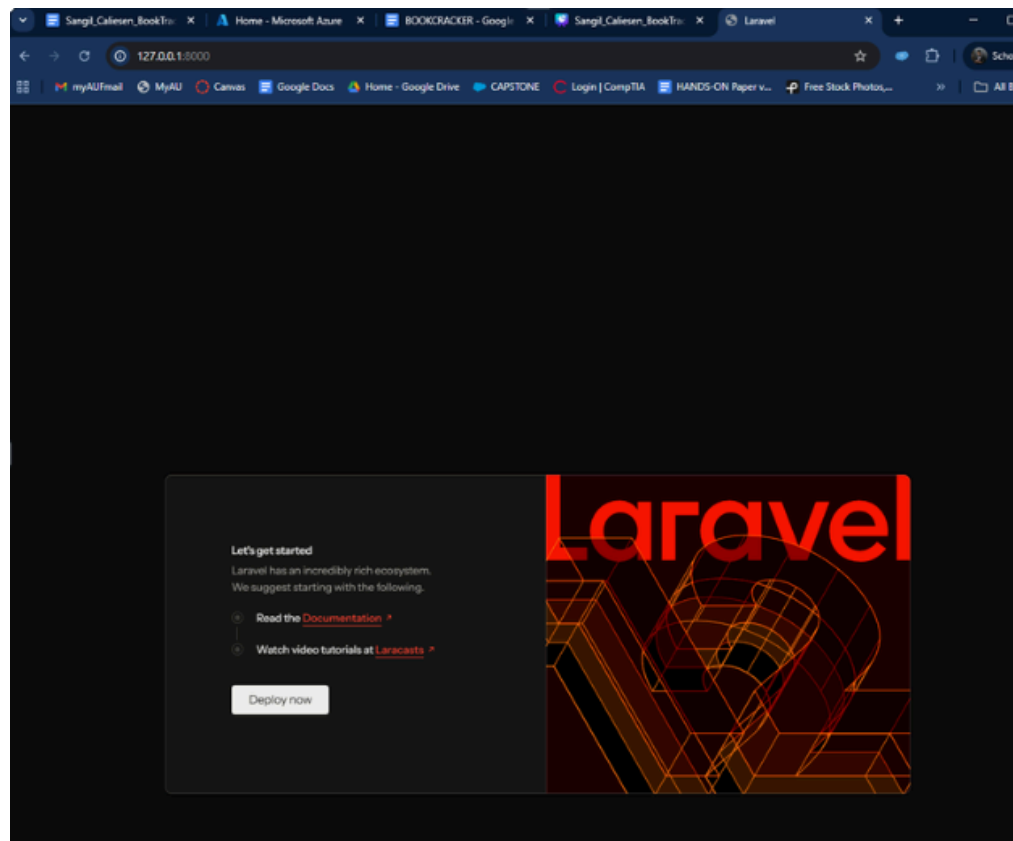
```
File Edit Selection View Go Run ... NatKim_BookTracker [Administrator]
EXPLORER
  NATKIM_BOOKTRACKER
    app
      Http
      Models
        Book.php
        User.php
      Providers
      bootstrap
      config
      database
      factories
      migrations
        0001_01_01_000000_create_users_table.php
        0001_01_01_000001_create_cache_table.p...
        0001_01_01_000002_create_jobs_table.php
        create_books_table.php
      seeders
      .gitignore
      database.sqlite
      public
      resources
      css
      js
      views
        index.blade.php
        welcome.blade.php
      routes
      storage
      tests
      vendor
      .editorconfig
      .env
      .env.example
      .gitattributes
      .gitignore
      artisan
      composer.json
      composer.lock
      package.json
      phpunit.xml
      README.md
      vite.config.js
  BookController.php 1
  Book.php
  welcome.blade.php 1
  index.blade.php 1

body.bg-gray-100:text-gray-800;font-sans > div.container.mx-auto.p-6 > div.grid.grid-cols-1.md:grid
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Azure Book Tracker</title>
7   <script src="https://cdn.tailwindcss.com"></script>
8   <link href="https://cdn.jsdelivr.net/npm/tailwindcss@2.2.19/dist/tailwind.min.css"></link>
9 </head>
10 <body class="bg-gray-100 text-gray-800 font-sans">
11
12   <nav class="bg-blue-700 text-white p-4 shadow-lg">
13     <div class="container mx-auto flex justify-between items-center">
14       <h1 class="text-2xl font-bold"><i class="fas fa-book-reader">
15       <span class="text-sm bg-blue-800 px-3 py-1 rounded">Deployed
16     </div>
17   </nav>
18
19   <div class="container mx-auto p-6">
20
21     @if(session('success'))
22       <div class="bg-green-100 border-1-4 border-green-500 text-gr
23       <p>{{ session('success') }}</p>
24     </div>
25   @endif
26
27   <div class="grid grid-cols-1 md:grid-cols-3 gap-6">
28
29     <div class="md:col-span-1">
30       <div class="bg-white rounded-lg shadow p-6">
31         <h2 class="text-xl font-bold mb-4">Add New Book</h2>
32         <form action="{{ route('books.store') }}" method="POST">
33           @csrf
34           <div class="mb-4">
35             <label class="block text-gray-700 text-sm fc
36             <input type="text" name="title" class="shad
37           </div>
38           <div class="mb-4">
39             <label class="block text-gray-700 text-sm fc
40             <input type="text" name="author" class="shac
41           </div>
42           <div class="mb-4">
43             <label class="block text-gray-700 text-sm fc
44             <select name="status" class="shadow border r
45               <option value="wishlist">Wishlist</optio
46               <option value="reading">Currently Readin
47               <option value="completed">Completed</opt
48             </select>
49           </div>
50           <div class="mb-4">
51             <label class="block text-gray-700 text-sm fc
52             <input type="file" name="cover_image" class=
```

5: Update routes

```
create_books_table.php BookController.php 1 Book.php index.blade.php web.php X
routes > web.php > ...
1  <?php
2
3  use Illuminate\Support\Facades\Route;
4  use App\Http\Controllers\BookController;
5  Route::get('/', [BookController::class, 'index'])->name('books.index');
6  Route::post('/book', [BookController::class, 'store'])->name('books.store');
7  Route::put('/book/{book}', [BookController::class, 'update'])->name('books.update');
8  Route::delete('/book/{book}', [BookController::class, 'destroy'])->name('books.destroy');
9
10 Route::get('/', function () {
11     return view('welcome');
12 });
13
```

6: Verify php artisan serve



7: Create Azure Server

Flexible server

Microsoft

Basics

Networking

Additional configuration

Tags

Review + create

Creation time

Estimated Server Creation Time (in minutes) 5

Product details

Azure Database for MySQL
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Basics (Change)

Subscription

Azure subscription 1

Resource group

BookTrackerRG

Server name

booktracker-server

Administrator login

kim

Location

Southeast Asia

Availability zone

No preference

High availability

Not enabled

MySQL version

8.0

Compute + storage

Burstable, 81ms, 1 vCores, 2 GiB RAM, 20 storage, Auto scale IOPS

Backup retention period (in days)

7 day(s)

Storage autogrow

Enabled

Geo-redundancy

Not enabled

Zonal Resiliency

No

Networking (Change)

Connectivity method

Public access (allowed IP addresses) and Private endpoint

Allow public access to this resource through the internet using a public IP address

Yes

Allow public access from any Azure service within Azure to this server

Yes

Firewall rules

1

SSL/TLS

SSL is enforced and TLS version is 1.2. This can be changed after server is created. [Learn more](#)

Additional configuration (Change)

Data encryption

Service-managed key

Server Parameter - lower_case_table_names

1

Estimated costs

Compute

USD 18.96/month

Free upto 750 hours

Standard, 81ms (1 vCore)

18.96

Storage

USD 2.76/month

Free upto 32 GiB

20 GiB @USD 0.14 per GiB

20 x 0.14

Auto scale IOPS

Auto scale IOPS is billed on usage in per million request increments. [Learn more](#)

Backup Retention

Backup retention is billed based on additional storage used for retaining backups. [Learn more](#)

Bandwidth

Outbound data transfer across services in different regions will incur additional charges. Any inbound data transfer is free. [Learn more](#)

Estimated total

USD 21.74/month

Charges will apply if you use above the free monthly limits. Please check [your usage](#) of free services. Final charges will appear in your local currency.

8: Create Web App

Create Web App ...

- Basics
- Database
- Deployment
- Networking
- Monitor + secure
- Tags
- Review + create

Summary

 **Web App**
by Microsoft

 Basic authentication for this app is currently disabled and may impact deployments. [Click to learn more.](#)

Details

Subscription	c27d724b-2e90-4f60-82d9-d67f89c0546c
Resource Group	BookTrackerRG
Name	natkimbooktracker
Secure unique default hostname	Enabled
Publish	Code
Runtime stack	PHP 8.2

App Service Plan

Name	ASP-kimnatrg-ba16
Operating System	Linux
Region	Southeast Asia
SKU	Basic
Size	Small
ACU	100 total ACU
Memory	1.75 GB memory

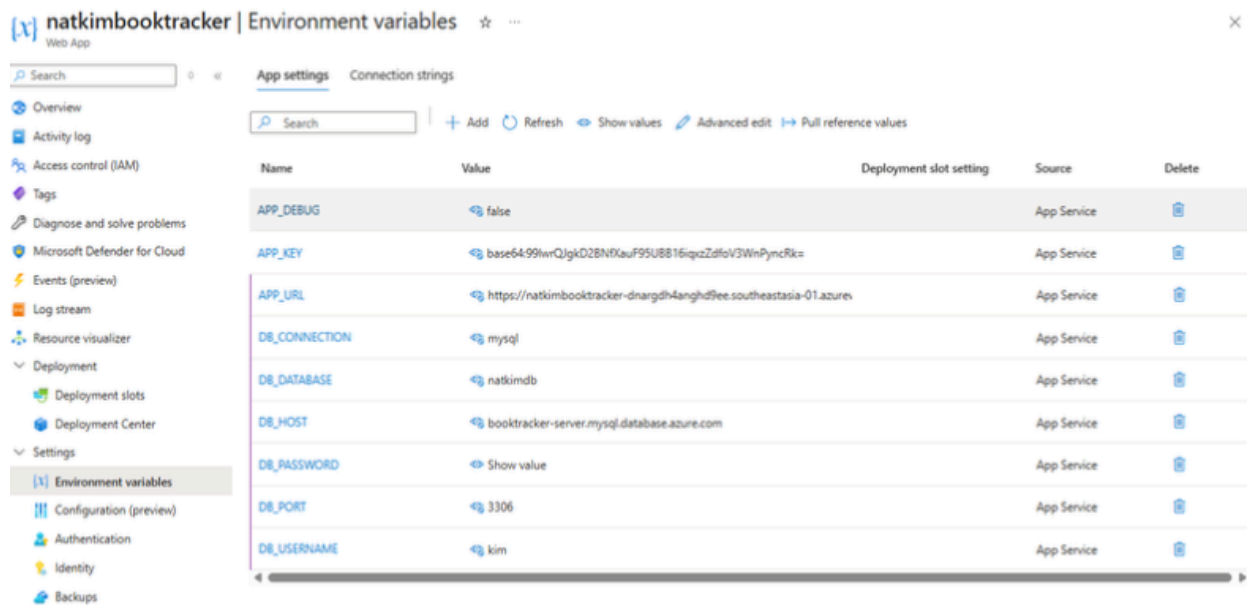
Monitor + secure

Application Insights	Not enabled
----------------------	-------------

Deployment

Basic authentication	Disabled
Continuous deployment	Not enabled / Set up after app creation

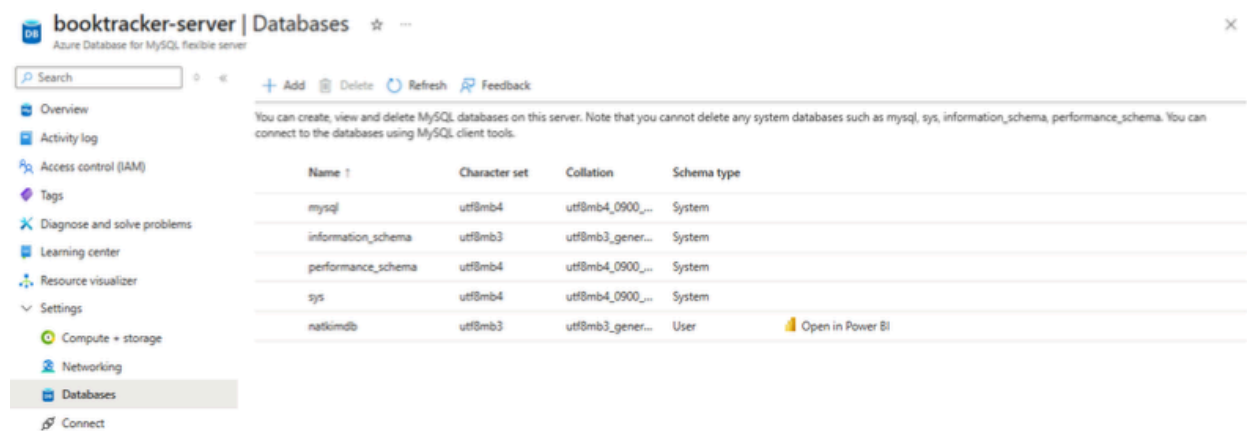
9: Update Environment Variables



The screenshot shows the 'Environment variables' page for the 'natkimbooktracker' web app. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), Log stream, Resource visualizer, Deployment, Deployment slots, Deployment Center, Settings, Environment variables (selected), Configuration (preview), Authentication, Identity, and Backups. The main content area has a search bar and tabs for 'App settings' and 'Connection strings'. Below the tabs, there are buttons for '+ Add', 'Refresh', 'Show values', 'Advanced edit', and 'Pull reference values'. A table lists the environment variables:

Name	Value	Deployment slot setting	Source	Delete
APP_DEBUG	false		App Service	
APP_KEY	base64:99hrrQJgkD2BNFXauF95U8816iqgzZdfcV3WnPyncRk=		App Service	
APP_URL	https://natkimbooktracker-dnargdh4anghd9ee.southeastasia-01.azure		App Service	
DB_CONNECTION	mysql		App Service	
DB_DATABASE	natkimdb		App Service	
DB_HOST	booktracker-server.mysql.database.azure.com		App Service	
DB_PASSWORD	Show value		App Service	
DB_PORT	3306		App Service	
DB_USERNAME	kim		App Service	

10: Create Database

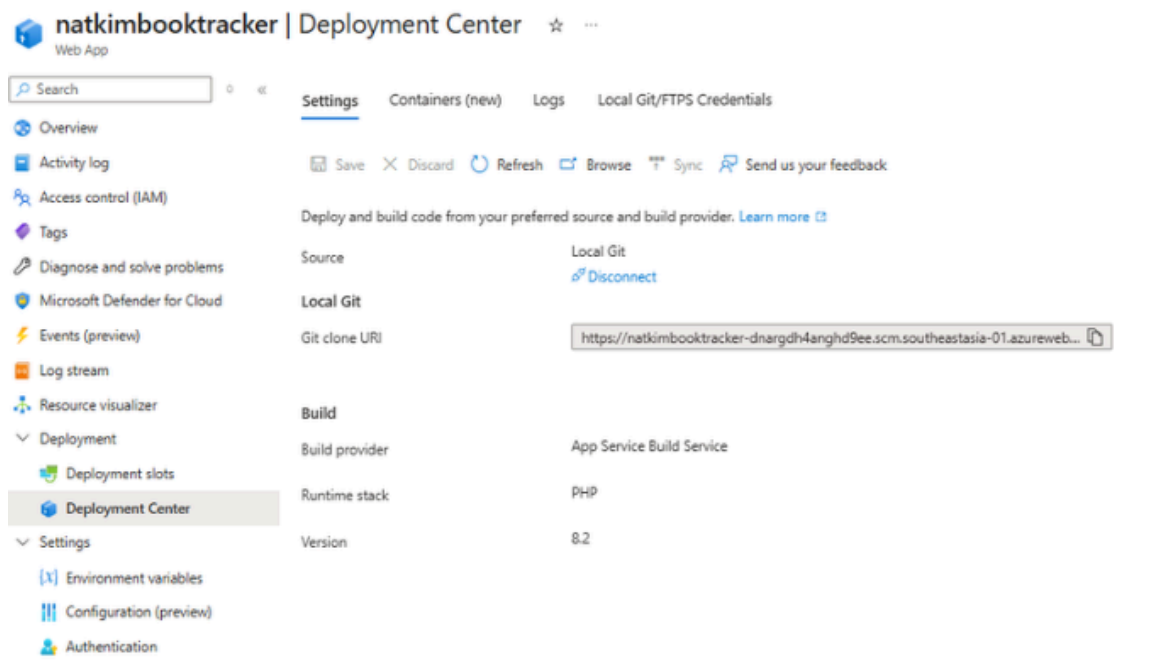


The screenshot shows the 'Databases' page for the 'booktracker-server' Azure Database for MySQL flexible server. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Learning center, Resource visualizer, Settings, Compute + storage (selected), Networking, Databases (selected), and Connect. The main content area has a search bar and buttons for '+ Add', 'Delete', 'Refresh', and 'Feedback'. Below the buttons, there is a message: 'You can create, view and delete MySQL databases on this server. Note that you cannot delete any system databases such as mysql, sys, information_schema, performance_schema. You can connect to the databases using MySQL client tools.' A table lists the databases:

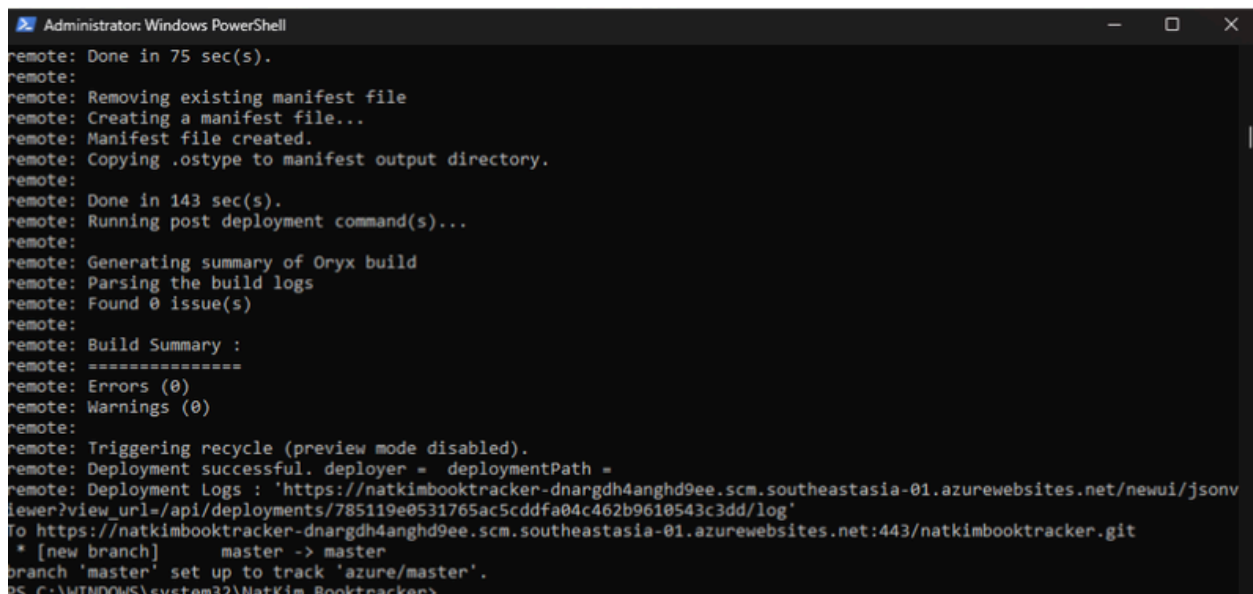
Name ↑	Character set	Collation	Schema type
mysql	utf8mb4	utf8mb4_0900_...	System
information_schema	utf8mb3	utf8mb3_gener...	System
performance_schema	utf8mb4	utf8mb4_0900_...	System
sys	utf8mb4	utf8mb4_0900_...	System
natkimdb	utf8mb3	utf8mb3_gener...	User

There is a link 'Open in Power BI' next to the 'natkimdb' row.

11: Activate Local Git as Source



12: git push to Azure



13: Turn OFF require_secure_transport

Top Modified Read-Only All

Top tab displays commonly modified parameters. You can hover over the info icon to get details of available parameters, navigate to **All** tab [Learn more](#)

Parameter name	Value	Parameter type
require_secure_transport	OFF	Dynamic

Showing 1 - 1 of 1 results.

14: php artisan migrate in SSH

```
Last login: Wed Nov 19 12:43:20 2025 from 169.254.130.3
```

```
      _____  
     /         \    _   _   _  
    /            \ /_   _/_/  
   /              \| | | | |  
  /                \| |_|_|  
 /_____\_____/\_____\_____  
|  _  |  _  |  _  |  _  |  _  | | | | | | |
| |_) | |_) | |_) | |_) | |_) |  
|___| |___| |___| |___| |___| |___|  
APP SERVICE ON LINUX
```

```
Documentation : http://aka.ms/webapp-linux  
PHP quickstart: https://aka.ms/php-qs  
PHP version   : 8.2.29  
Instance Name : lw0adlw0000XK  
Instance Id   : b1b997ff8f7bd42ca2bc93b11283ddcc908e8ad9ed4443c0e02f8f2c5176ec22
```

```
Note: Any data outside '/home' is not persisted  
root@51980874a8a8:/home# cd /home/site/wwwroot  
root@51980874a8a8:/home/site/wwwroot# php artisan migrate --force
```

```
INFO Preparing database.
```

```
Creating migration table ..... 445.17ms DONE
```

```
INFO Running migrations.
```

```
0001_01_01_000000_create_users_table ..... 601.47ms DONE  
0001_01_01_000001_create_cache_table ..... 142.82ms DONE  
0001_01_01_000002_create_jobs_table ..... 272.91ms DONE  
create_books_table ..... 113.70ms DONE
```

```
root@51980874a8a8:/home/site/wwwroot#
```

How we maximized Microsoft Azure for our Project:

1. **Azure App Service** (PaaS for the Web App)
 - a. Zero Server Management - by using Azure App Service on Linux for our PHP app, we have successfully eliminated the need for provision, patch or maintain a Virtual Machine, due to Azure being able to handle the Operating System (OS), PHP runtime, web server (Apache/Nginx), and all security necessities.
 - b. Built-in Diagnostics and SSH - We were able to utilize Kudu and the built-in SSH console in inspecting the file paths and configurations real-time.
2. **Azure Database for MySQL** (PaaS for the Database)
 - a. Managed Database Service - Instead of setting up MySQL on a virtual machine, we utilized Azure Database for MySQL (Flexible Server).